

ECX2400 - ECX5400: Design and Evaluation of Economic, Public, and Social Programs

Take Home Final Exam, 2026, Rigissa Megalokonomou

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MARKS ALLOCATED TO QUESTIONS WITHIN THIS ASSESSMENT TASK

Question	1	2	Penalty	TOTAL
Allocated Marks	50	50		100
OFFICIAL USE ONLY				
Mark received				
Second marking				

Question 1

Part(1)

1. Stata command used

The Stata commands used to draw the required i.i.d. sample and estimate the OLS regression with robust standard errors were:

```
use wage_exam_V2.dta, clear
set seed 35954396
bsample 350

reg lwage exper educ tenure, robust
```

2. Coefficient and statistical significance

From the stata output

```
. * Estimate the OLS wage equation with robust standard errors
. reg lwage exper educ tenure, robust
```

Linear regression		Number of obs	=	350
		F(3, 346)	=	29.59
		Prob > F	=	0.0000
		R-squared	=	0.1802
		Root MSE	=	.40169

lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
exper	.0182656	.0054975	3.32	0.001	.0074528	.0290784
educ	.0643004	.0096086	6.69	0.000	.0454017	.0831991
tenure	.023725	.0043895	5.40	0.000	.0150916	.0323584
_cons	5.478104	.1620783	33.80	0.000	5.159322	5.796887

The estimated OLS wage equation is

$$\widehat{\log(wage)} = \underset{(0.1621)}{5.4781} + \underset{(0.0055)}{0.0183} \text{exper} + \underset{(0.0096)}{0.0643} \text{educ} + \underset{(0.0044)}{0.0237} \text{tenure}$$

The estimated coefficient on *exper* is 0.0183, with a robust standard error of 0.0055. The reported *t*-statistic is 3.32, and the *p*-value is 0.001. Since $0.001 < 0.05$, I reject $H_0 : \beta_{\text{exper}} = 0$, so *exper* is statistically significant at the 5% level.

3. Economic interpretation

A one-year increase in labor market experience is associated with a 1.83% increase in monthly wages, holding education and tenure constant.

4. Economic size and statistical precision

Although the estimated return is statistically precise, it is not economically large. It is statistically precise because the 95% confidence interval [0.00745, 0.02908] is entirely positive and $p=0.001$. A 1.83% increase per year is modest (as typical experience returns in the 1–3% range are not considered large), but it still can accumulate over a career.

5. Causal interpretation

No, this estimate should not necessarily be interpreted as causal. Labour market experience is not randomly assigned, and workers with different experience levels may also differ in unobserved characteristics such as ability, motivation, occupation, or family background. If these omitted factors affect wages and are correlated with *exper*, then $E(u \mid \text{exper}, \text{educ}, \text{tenure}) = 0$ is violated and the OLS coefficient should be interpreted as a conditional association rather than a causal effect.

Part (2)

1. Hypotheses

The researcher's claim is that the coefficient on education equals the coefficient on tenure. Thus, the null and alternative hypotheses are

$$H_0 : \beta_2 = \beta_3$$

$$H_1 : \beta_2 \neq \beta_3$$

Equivalently,

$$H_0 : \beta_2 - \beta_3 = 0, \quad H_1 : \beta_2 - \beta_3 \neq 0$$

2. Hypothesis test

To conduct the test using a *t*-test, I rewrote the model by defining

$$\text{sum_ed_ten} = \text{educ} + \text{tenure}$$

and

$$\text{diff_ed_ten} = \text{educ} - \text{tenure}.$$

The Stata commands used were:

```
gen sum_ed_ten = educ + tenure
gen diff_ed_ten = educ - tenure

reg lwage exper sum_ed_ten diff_ed_ten, robust
```

In this reparameterised model, the coefficient on *diff_ed_ten* equals $(\beta_{\text{educ}} - \beta_{\text{tenure}})/2$, so testing $\beta_{\text{educ}} = \beta_{\text{tenure}}$ is equivalent to testing whether the coefficient on *diff_ed_ten* is equal to zero.

3. Test statistic and p-value

From the robust regression, the estimated coefficient on *diff_ed_ten* is 0.0203, with a robust standard error of 0.0055. The reported *t*-statistic is 3.71, and the *p*-value is 0.000.

```

. * Test H0: beta_educ = beta_tenure
. *t-test
. gen sum_ed_ten = educ + tenure

. gen diff_ed_ten = educ - tenure

.
. reg lwage exper sum_ed_ten diff_ed_ten, robust

```

Linear regression

Number of obs	=	350
F(3, 346)	=	29.59
Prob > F	=	0.0000
R-squared	=	0.1802
Root MSE	=	.40169

lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
exper	.0182656	.0054975	3.32	0.001	.0074528	.0290784
sum_ed_ten	.0440127	.0050934	8.64	0.000	.0339947	.0540307
diff_ed_ten	.0202877	.0054638	3.71	0.000	.0095412	.0310342
_cons	5.478104	.1620783	33.80	0.000	5.159322	5.796887

4. Conclusion at the 5% significance level

Since the p -value is less than 0.05, I reject the null hypothesis at the 5% significance level. Therefore, the data provide evidence that the effect of one additional year of education on log wages is statistically different from the effect of one additional year of tenure.

5. Whether the data support the researcher's claim

The data do not support the researcher's claim. In the original regression, the estimated coefficient on education is 0.0643, while the estimated coefficient on tenure is 0.0237. Since the t -test rejects $H_0 : \beta_{educ} = \beta_{tenure}$, the estimated effects are statistically different.

Part (3)

1. Estimated coefficient on educ

```

. * Original regression from Part (1)
. reg lwage exper educ tenure, robust

```

Linear regression

Number of obs	=	350
F(3, 346)	=	29.59
Prob > F	=	0.0000
R-squared	=	0.1802
Root MSE	=	.40169

lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
exper	.0182656	.0054975	3.32	0.001	.0074528	.0290784
educ	.0643004	.0096086	6.69	0.000	.0454017	.0831991
tenure	.023725	.0043895	5.40	0.000	.0150916	.0323584
_cons	5.478104	.1620783	33.80	0.000	5.159322	5.796887

```

.
. * Re-estimate excluding exper
. reg lwage educ tenure, robust

```

Linear regression

Number of obs	=	350
F(2, 347)	=	30.70
Prob > F	=	0.0000
R-squared	=	0.1558
Root MSE	=	.40704

lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
educ	.0484341	.0090556	5.35	0.000	.0306233	.0662448
tenure	.0256957	.0042577	6.04	0.000	.0173216	.0340699
_cons	5.88936	.1278844	46.05	0.000	5.637834	6.140886

In the original regression from Part (1), the estimated coefficient on educ is

$$\hat{\beta}_{educ}^{original} = 0.0643$$

After excluding `exper` from the model, the estimated coefficient on educ becomes

$$\hat{\beta}_{educ}^{no\ exper} = 0.0484$$

2. Change in the estimated return to education

The estimated return to education decreases after excluding `exper`. The change is

$$0.0484 - 0.0643 = -0.0159$$

Therefore, excluding `exper` reduces the estimated education coefficient by about 0.0159 log points, or approximately 1.59 percentage points. In percentage terms, this is a decrease of approximately

$$\frac{0.0484 - 0.0643}{0.0643} \times 100 \approx -24.7\%$$

3. Why excluding `exper` may change the coefficient on educ

Excluding `exper` may change the educ coefficient because experience affects wages and is correlated with education. In my estimates, omitting `exper` causes the educ coefficient to fall from 0.0643 to 0.0484. This negative change reflects omitted variable bias, suggesting a negative correlation between educ and `exper` in my sample.

4. Implication if the coefficient changes very little

If the coefficient on education changed very little after excluding `exper`, it would suggest that education and labour market experience are nearly uncorrelated (conditional on tenure), or that experience has a very small effect on wages, or that omitted variable bias is small. In my case, however, the coefficient changed substantially (a 24.7% decrease), so this situation does not apply.

5. Whether omitted variable bias appears quantitatively important

Yes, in this setting, omitted variable bias seems quantitatively significant. Excluding `exper` reduces the educ coefficient from 0.0643 to 0.0484, a 24.7% drop (0.0159). This represents a meaningful 1.6 percentage point change in the estimated return to education. Thus, excluding labour market experience substantially alters the estimated return to education.

Part (4)

1. Omitted variable

The student's conclusion is not necessarily correct. One possible omitted variable is whether a worker is in a higher-paying or senior occupation type. This omitted variable may directly affect wages because workers in senior, professional, or higher-responsibility jobs usually earn higher wages, even after controlling for education and tenure.

2. Correlation with *exper*

Being in a higher-paying or senior occupation type is likely to be positively correlated with labour market experience. Workers with more experience are more likely to move out of entry-level or lower-paying jobs and into senior positions or higher-paying occupations. Therefore, this omitted variable is likely to be positively related to *exper*.

3. Direction of omitted variable bias

The resulting omitted variable bias is likely to be upward. Higher-paying or senior occupation types have a positive effect on wages and are also positively correlated with *exper*. If this variable is omitted, part of the wage premium from better occupations may be incorrectly attributed to labour market experience, causing the coefficient on *exper* to overstate the true causal return to experience.

4. Why statistical significance is insufficient for causality

Statistical significance only shows that *exper* is conditionally associated with log wages in this regression. It does not prove that labour market experience causes higher wages, because *exper* is not randomly assigned. If omitted variables such as higher-paying or senior occupation type are included in the error term and correlated with *exper*, then the zero conditional mean assumption, $E(u \mid \text{exper}, \text{educ}, \text{tenure}) = 0$, is violated.

5. Least convincing result from a causal perspective

The least convincing empirical result from a causal perspective is the interpretation of the positive and statistically significant coefficient on *exper* in Part (1) as a causal return to labour market experience. Although the coefficient on *exper* is statistically significant, the regression only controls for education and tenure. Since important wage determinants such as higher-paying or senior occupation type are omitted, the coefficient on *exper* is better interpreted as a conditional association rather than a causal effect.

Part (5)

1. Running variable, cutoff, and treatment variable

The running variable is years of education centred at the cutoff:

$$r_i = \text{educ}_i - 14.$$

The cutoff is $\text{educ}_i = 14$, equivalently $r_i = 0$. The treatment variable is

$$D_i = 1(\text{educ}_i \geq 14),$$

which equals one if the individual has at least 14 years of education, and zero otherwise.

2. Stata commands

The Stata commands used to construct r_i and D_i , and to estimate the RDD-style regression with robust standard errors, were:

```

gen running = educ - 14
gen college = (educ >= 14)

reg lwage college running, robust

```

3. Estimated coefficient on D_i

```

. gen running = educ - 14
. gen college = (educ >= 14)
.
. reg lwage college running, robust

```

Linear regression		Number of obs	=	350
		F(2, 347)	=	15.73
		Prob > F	=	0.0000
		R-squared	=	0.0730
		Root MSE	=	.42651

lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
college	.0618173	.0986577	0.63	0.531	-.1322252	.2558597
running	.0397713	.019374	2.05	0.041	.001666	.0778767
_cons	6.716202	.0550574	121.99	0.000	6.607913	6.82449

The estimated coefficient on D_i , reported as college in the regression, is 0.0618, with a robust standard error of 0.0987. The t -statistic is 0.63, and the p -value is 0.531, so the coefficient is not statistically significant at the 5% level. Interpreted in words, individuals just at or above the 14-year education cutoff have about 6.18% higher wages than individuals just below the cutoff, after controlling linearly for the running variable, but this estimated jump is statistically imprecise.

4. Why individuals with 13 and 14 years may be more comparable

Individuals with 13 and 14 years of education may be more comparable because they are close to the cutoff and differ by only one year of education. In contrast, individuals with 10 and 14 years of education may differ much more in ability, family background, motivation, and labour market opportunities. The RDD idea relies on comparisons close to the cutoff being more like local comparisons.

5. Key assumption for causal interpretation

The key assumption is that all other factors affecting wages change smoothly at the cutoff. In other words, apart from the treatment indicator D_i , there should be no discontinuous jump in observed or unobserved wage determinants at $educ_i = 14$. Under this continuity assumption, the jump in log wages at the cutoff can be interpreted as the causal effect of completing college for individuals near the cutoff.

6. Why 13-year and 14-year individuals may still differ

Individuals with 13 and 14 years of education may still differ systematically because education is chosen rather than randomly assigned. For example, people who complete at least 14 years may have higher ability, stronger motivation, better family background, or better access to education. Therefore, differences in wages near the cutoff may reflect selection, not only the causal effect of completing college.

7. Limitation of using years of education as the running variable

Years of education is not a fully convincing running variable in this context because individuals can choose how much education to obtain. The cutoff at 14 years is not clearly generated by an external rule that assigns treatment as good as randomly near the cutoff. Therefore, this proposed RDD may suffer from selection around the cutoff, so the estimated coefficient on college should be interpreted cautiously rather than as strong causal evidence.

Question 2: Panel Data, Pooled OLS, and Heterogeneity Bias

Part (a)

1. Panel identifiers, N , and T

The Stata commands used to inspect and declare the panel dataset were:

```
use SIMUDATA_V2.dta, clear

describe
summarize

sort id t
xtset id t

tab t
```

The individual identifier is id , and the time identifier is t . From the Stata output, id ranges from 1 to 100, so the number of individuals is $N = 100$. The time variable t ranges from 1 to 5, so the number of time periods is $T = 5$, and the total number of observations is $5 \times 100 = 500$.

2. Long form or wide form

The dataset is stored in long form. This is because each row represents one individual-time observation, identified by a unique pair (id, t) . The command `isid id t` confirms that id and t uniquely identify each observation. The data have 500 rows, which equals 100 individuals observed over 5 time periods.

3. Balanced or unbalanced panel

The panel is balanced. The `xtset id t` output reports that the panel is strongly balanced. This is also consistent with `tab t`, which shows 100 observations in each time period from $t = 1$ to $t = 5$.

4. Empirical problem in an unbalanced panel

One empirical problem in an unbalanced panel is non-random attrition or missing observations. If individuals drop out of the panel for reasons related to the outcome variable or explanatory variables, the remaining sample may not be representative. This can lead to sample selection bias and biased coefficient estimates.

Part (b)

1. Composite error term

The error term v_{it} is called a composite error term because it contains two components:

$$v_{it} = c_i + u_{it}$$

The first component, c_i , captures time-invariant unobserved individual characteristics, while u_{it} captures idiosyncratic shocks that vary across individuals and time. Hence it is a composite of individual-specific and random noise.

2. Dependence over time for the same individual

Observations for the same individual may not be independent over time because they share the same unobserved individual effect c_i . For example, if an individual has high unobserved ability, this affects their outcomes in every period. Therefore, the composite errors v_{it} and v_{is} for the same individual are likely to be correlated across time.

3. Condition for pooled OLS to be unbiased

Pooled OLS can produce unbiased estimates of γ_2 only if the composite error term v_{it} is uncorrelated with x_{it} . Since $v_{it} = c_i + u_{it}$, this requires both $Cov(x_{it}, c_i) = 0$ and $Cov(x_{it}, u_{it}) = 0$. If x_{it} is correlated with the unobserved individual effect c_i , pooled OLS suffers from heterogeneity bias.

4. Example where c_i may be correlated with x_{it}

Suppose y_{it} is wages and x_{it} is labour market experience or training participation. The unobserved individual effect c_i may represent ability or motivation. More able or more motivated workers may both earn higher wages and accumulate more experience or be more likely to participate in training, so c_i may be correlated with x_{it} .

5. Why following the same individuals over time helps

Following the same individuals over time helps because panel data allow us to separate time-invariant unobserved heterogeneity c_i from time-varying shocks u_{it} . This helps us think about methods that remove or control for c_i , such as first differences or fixed effects. In a pooled cross section, we usually observe different individuals in different periods, so we cannot use within-individual changes to control for fixed individual characteristics.

Part (c)

1. Stata command used

The Stata command used to estimate the pooled OLS model was:

```
reg y x1 x2 x3 x4
```

This gives 500 observations in the regression.

2. Estimated coefficient on $x_{it,2}$, standard error, and statistical significance

```
. * Pooled OLS
. reg y x1 x2 x3 x4
```

Source	SS	df	MS	Number of obs	=	500
Model	10877.716	4	2719.429	F(4, 495)	=	935.90
Residual	1438.31127	495	2.90567933	Prob > F	=	0.0000
				R-squared	=	0.8832
				Adj R-squared	=	0.8823
Total	12316.0273	499	24.6814174	Root MSE	=	1.7046

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	2.488326	.0416142	59.80	0.000	2.406564	2.570088
x2	-1.920762	.1148951	-16.72	0.000	-2.146504	-1.69502
x3	.0071604	.2845003	0.03	0.980	-.5518167	.5661374
x4	-.1522468	.4177954	-0.36	0.716	-.9731178	.6686241
_cons	2.129437	.1889854	11.27	0.000	1.758124	2.500749

The estimated pooled OLS equation is

$$\hat{y}_{it} = 2.1294 + 2.4883x_{it,1} - 1.9208x_{it,2} + 0.0072x_{it,3} - 0.1522x_{it,4}.$$

(0.1890) (0.0416) (0.1149) (0.2845) (0.4178)

The estimated coefficient on $x_{it,2}$ is -1.9208 , with a standard error of 0.1149 . The reported t -statistic is -16.72 , and the p -value is 0.000 . Since $0.000 < 0.05$, the coefficient on $x_{it,2}$ is statistically significant at the 5% level.

3. Interpretation of the coefficient on $x_{it,2}$

Holding $x_{it,1}$, $x_{it,3}$, and $x_{it,4}$ constant, a one-unit increase in $x_{it,2}$ is associated with a decrease in y_{it} of about 1.9208 units. This is a conditional association from the pooled OLS regression.

4. Causal interpretation

The estimated coefficient on $x_{it,2}$ should not necessarily be interpreted causally. A key panel data issue is unobserved time-invariant heterogeneity, c_i , such as ability, motivation, or individual background. If c_i is correlated with $x_{it,2}$, then the pooled OLS error term contains c_i , so $E(v_{it} | x_{it,2}) \neq 0$. In that case, the pooled OLS estimate suffers from heterogeneity bias and should be interpreted as an association rather than a causal effect.

Part (d)

1. Stata command used

The Stata commands used to generate the time dummies and estimate the pooled OLS model were:

```
gen d1 = (t == 1)
gen d2 = (t == 2)
gen d3 = (t == 3)
gen d4 = (t == 4)

reg y d1 d2 d3 d4 x1 x2 x3 x4
```

The final time period, $t = 5$, is omitted as the base period.

2. Number of time dummies

There are $T = 5$ time periods in the data, so the model includes $T - 1 = 4$ time dummies. These are $d1, d2, d3, d4$, corresponding to $t = 1, 2, 3, 4$. The dummy for $t = 5$ is omitted as the base period to avoid perfect multicollinearity.

3. One estimated time dummy coefficient and standard error

```

. * Pooled OLS with time dummies
. * Final time period t = 5 is the omitted base period
.
. gen d1 = (t == 1)
.
. gen d2 = (t == 2)
.
. gen d3 = (t == 3)
.
. gen d4 = (t == 4)
.
.
. reg y d1 d2 d3 d4 x1 x2 x3 x4

```

Source	SS	df	MS	Number of obs	=	500
Model	11046.5356	8	1380.81694	F(8, 491)	=	534.06
Residual	1269.49173	491	2.58552286	Prob > F	=	0.0000
				R-squared	=	0.8969
				Adj R-squared	=	0.8952
Total	12316.0273	499	24.6814174	Root MSE	=	1.608

	y	Coefficient	Std. err.	t	P> t	[95% conf. interval]
	d1	-1.013994	.2275548	-4.46	0.000	-1.461096 -.5668932
	d2	-.9006612	.2274274	-3.96	0.000	-1.347512 -.4538102
	d3	-.9044927	.2275513	-3.97	0.000	-1.351587 -.4573982
	d4	-1.834911	.2277361	-8.06	0.000	-2.282369 -1.387453
	x1	2.483958	.0392632	63.26	0.000	2.406814 2.561103
	x2	-1.930245	.1084443	-17.80	0.000	-2.143317 -1.717173
	x3	.0012955	.2683913	0.00	0.996	-.5260418 .5286328
	x4	-.0047256	.3959881	-0.01	0.990	-.7827658 .7733146
	_cons	3.005056	.2285906	13.15	0.000	2.55592 3.454193

Using $d1$ as an example, the estimated coefficient on $d1$ is -1.0140 , with a standard error of 0.2276 . The regression uses 500 observations.

4. Interpretation of the time dummy coefficient

The coefficient on $d1$ means that, holding $x1, x2, x3$, and $x4$ constant, the average value of y in period $t = 1$ is about 1.0140 units lower than in the omitted base period $t = 5$. This interpretation is relative to the final time period because $t = 5$ is the excluded category.

5. Why include time dummies

Panel data applications often include time dummies to control for common shocks or time effects that affect all individuals in the same period. For example, macroeconomic conditions, policy changes, or aggregate trends may shift y over time. Including time dummies helps separate these common time effects from the effects of $x1, x2, x3$, and $x4$.

Part (e)

1. Stata commands used

The Stata commands used for the two pooled OLS models were:

```
reg y x2
reg y x1 x2 x3 x4
```

Model 1 includes only $x_{it,2}$, while Model 2 adds $x_{it,1}$, $x_{it,3}$, and $x_{it,4}$ as additional controls.

2. Comparison of the estimated coefficient on $x_{it,2}$

. * Model 1						
. reg y x2						
Source	SS	df	MS	Number of obs	=	500
Model	422.201475	1	422.201475	F(1, 498)	=	17.68
Residual	11893.8258	498	23.8831843	Prob > F	=	0.0000
				R-squared	=	0.0343
				Adj R-squared	=	0.0323
Total	12316.0273	499	24.6814174	Root MSE	=	4.887
y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x2	-.6979179	.1659933	-4.20	0.000	-1.024051	-.3717844
_cons	2.000058	.2885135	6.93	0.000	1.433204	2.566912
. * Model 2						
. reg y x1 x2 x3 x4						
Source	SS	df	MS	Number of obs	=	500
Model	10877.716	4	2719.429	F(4, 495)	=	935.90
Residual	1438.31127	495	2.90567933	Prob > F	=	0.0000
				R-squared	=	0.8832
				Adj R-squared	=	0.8823
Total	12316.0273	499	24.6814174	Root MSE	=	1.7046
y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	2.488326	.0416142	59.80	0.000	2.406564	2.570088
x2	-1.920762	.1148951	-16.72	0.000	-2.146504	-1.69502
x3	.0071604	.2845003	0.03	0.980	-.5518167	.5661374
x4	-.1522468	.4177954	-0.36	0.716	-.9731178	.6686241
_cons	2.129437	.1889854	11.27	0.000	1.758124	2.500749

In Model 1, the estimated coefficient on $x_{it,2}$ is -0.6979 , with a standard error of 0.1660 , using 500 observations. In Model 2, the estimated coefficient on $x_{it,2}$ is -1.9208 , with a standard error of 0.1149 , also using 500 observations. After adding the additional controls, the coefficient becomes more negative by 1.2228 , changing from -0.6979 to -1.9208 .

3. Implication for omitted variable bias

The large change in the coefficient suggests that omitted variable bias may be important in Model 1. When $x_{it,1}$, $x_{it,3}$, and $x_{it,4}$ are omitted, the coefficient on $x_{it,2}$ may partly capture the effects of these omitted variables. Since adding the controls makes the coefficient on $x_{it,2}$ much more negative, Model 1 appears to understate the negative association between $x_{it,2}$ and y_{it} .

4. More credible specification

Model 2 is more credible than Model 1 because it controls for more relevant explanatory variables. By including $x_{it,1}$, $x_{it,3}$, and $x_{it,4}$, Model 2 is less likely to suffer from omitted variable bias from these controls. However, because this is still pooled OLS, it may still be biased if unobserved time-invariant individual heterogeneity c_i is correlated with $x_{it,2}$.

Appendix: Stata Do-file

```
* Reset
clear all

* Set working directory
cd "C:\Users\Matilda Fang\Desktop\2026 S1\EXC2400\Final exam"

* Create a log file
capture log close
log using "Assessment.log", replace

* Question 1
* Question 1 Part (1)
use wage_exam_V2.dta, clear

set seed 35954396
bsample 350
describe

* Estimate the OLS wage equation with robust standard errors
reg lwage exper educ tenure, robust

* Question 1 Part (2)
* Test  $H_0: \beta_{educ} = \beta_{tenure}$ 
*t-test
gen sum_ed_ten = educ + tenure
gen diff_ed_ten = educ - tenure

reg lwage exper sum_ed_ten diff_ed_ten, robust

* Question 1 Part (3)
* Original regression from Part (1)
reg lwage exper educ tenure, robust

* Re-estimate excluding exper
reg lwage educ tenure, robust

* Question 1 Part (5)
gen running = educ - 14
gen college = (educ >= 14)

reg lwage college running, robust
```

```

* Question 2
* Question 2 Part (a)
use SIMUDATA_V2.dta, clear

describe
summarize

sort id t
xtset id t

tab t

* Question 2 Part (c)
* Pooled OLS
reg y x1 x2 x3 x4

* Question 2 Part (d)
* Pooled OLS with time dummies
* Final time period t = 5 is the omitted base period
gen d1 = (t == 1)
gen d2 = (t == 2)
gen d3 = (t == 3)
gen d4 = (t == 4)

reg y d1 d2 d3 d4 x1 x2 x3 x4

* Question 2 Part (e)
* Model 1
reg y x2

* Model 2
reg y x1 x2 x3 x4

log close

```